

Hidden keys and repeated biometric records

Can multiple protected records reveal identity when keys remain hidden?

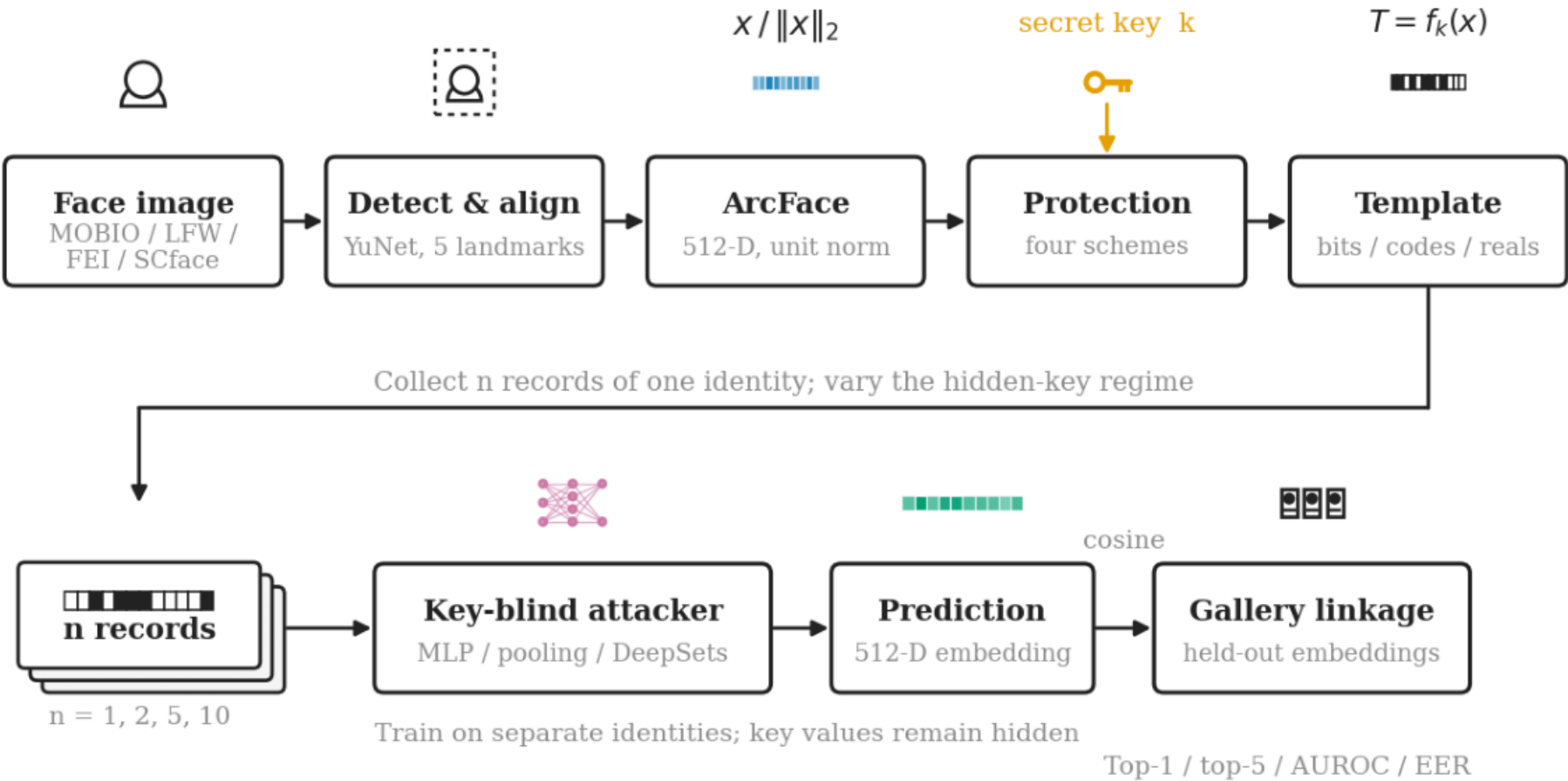
Fresh-key learned attacks: uncertainty remains; no equivalence claim.

Recurring transforms: large gains from multiple same-identity records.

Boundary location changes with the dataset and identity partition.

Independent study. No exact benchmark reproduction or universal privacy claim.

Experimental architecture



Train on separate identities. Reconstruct an embedding, then link to a held-out gallery.

Dataset and protection coverage

Dataset	Identities	Train/val/test	Embeddings	Variation
MOBIO	150	90/30/30	1,799 / 1,800	Sessions
LFW	125	75/25/25	1,500 / 1,500	In-the-wild
FEI	200	120/40/40	2,378 / 2,400	Pose/expression
SCface	130	78/26/26	2,851 / 2,860	Camera/distance

BioHash: 128 bits; four datasets; includes a Haar-sign control on MOBIO.

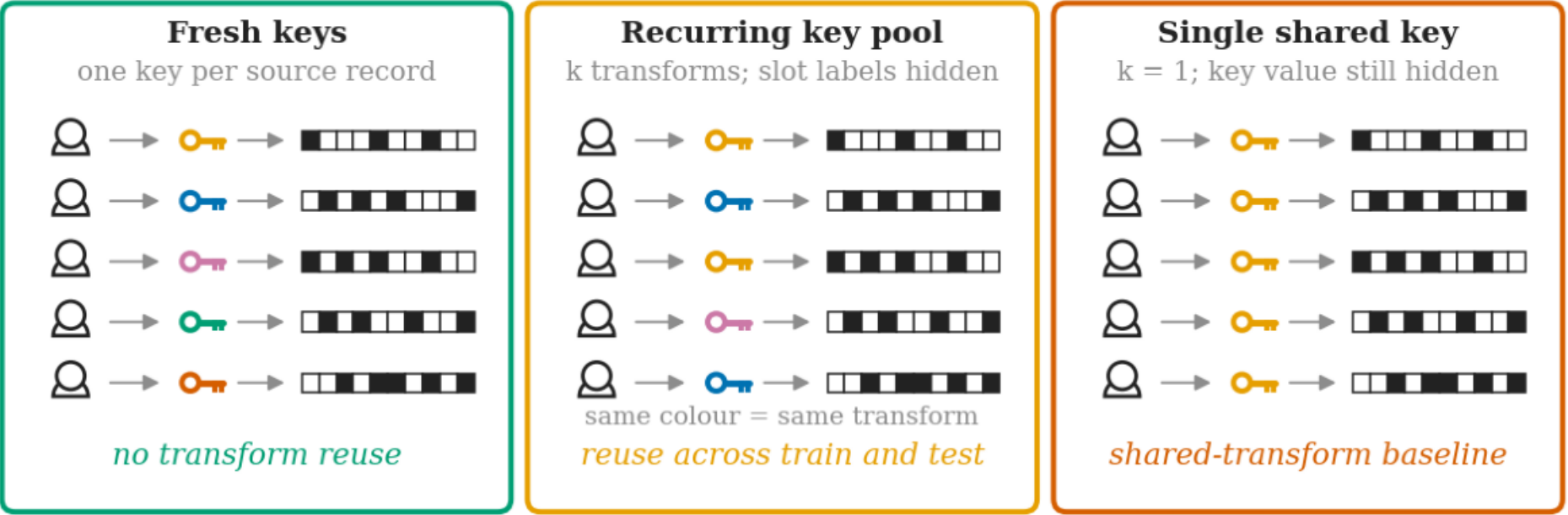
MLP-Hash: 512 bits; MOBIO; paper-specified, not source-exact.

IoM-GRP: 300 categorical codes (q=16); MOBIO/FEI/SCface pilots.

PolyProtect: 170 real values (m=5, overlap=2); MOBIO/FEI/SCface pilots.

SCface: mugshot gallery and visible surveillance probes; 9 detection failures, all identities eligible.

What changes between key regimes?

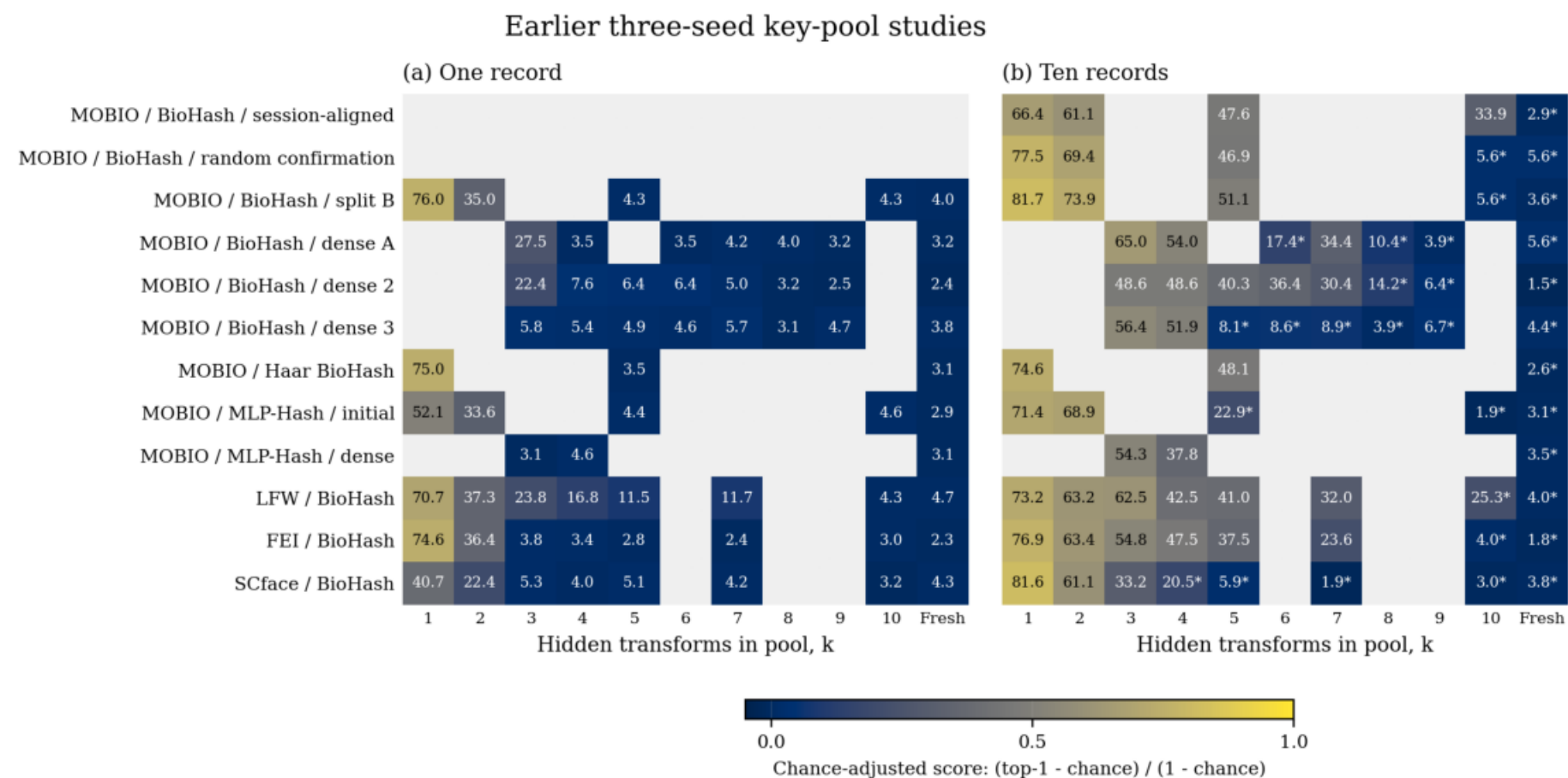


Learned attacks inconclusive **Multi-record gains observed** **Strong linkage observed**
PolyProtect outside the theorem boundary depends on dataset and split hidden does not mean unlearnable

Protocol Disjoint train / validation / test identities; one held-out gallery image per identity
8 sets per identity; 3 seeds earlier, 1 in pilots; identity-clustered intervals

Key values stay hidden in every regime. Recurring transforms are shared across identity splits.

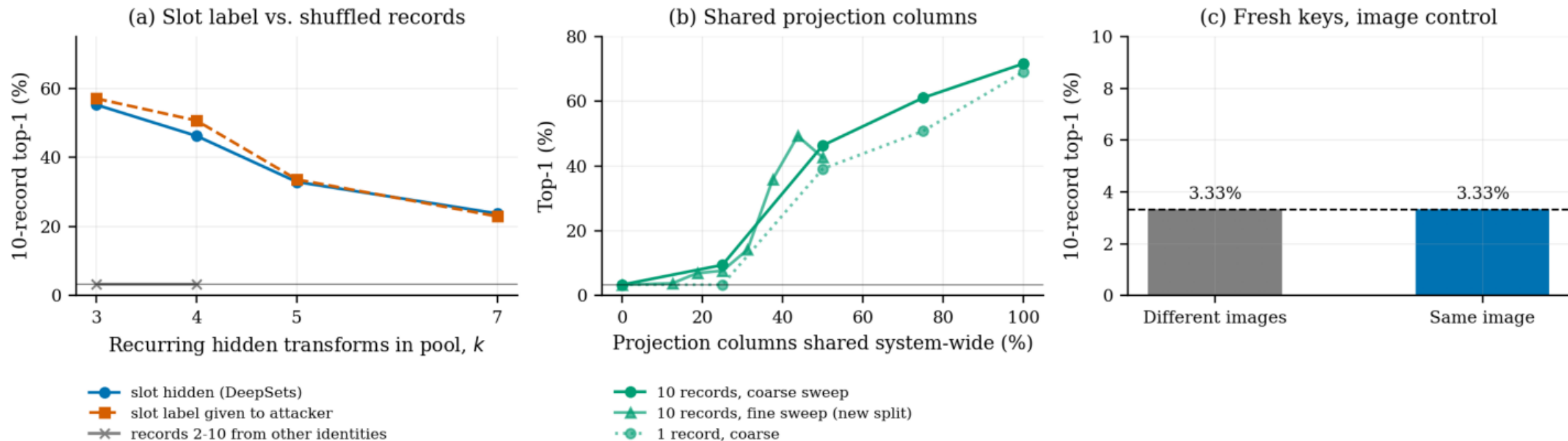
Earlier three-seed key-pool studies



73 conditions from 12 earlier studies. New one-seed pilots are reported separately.

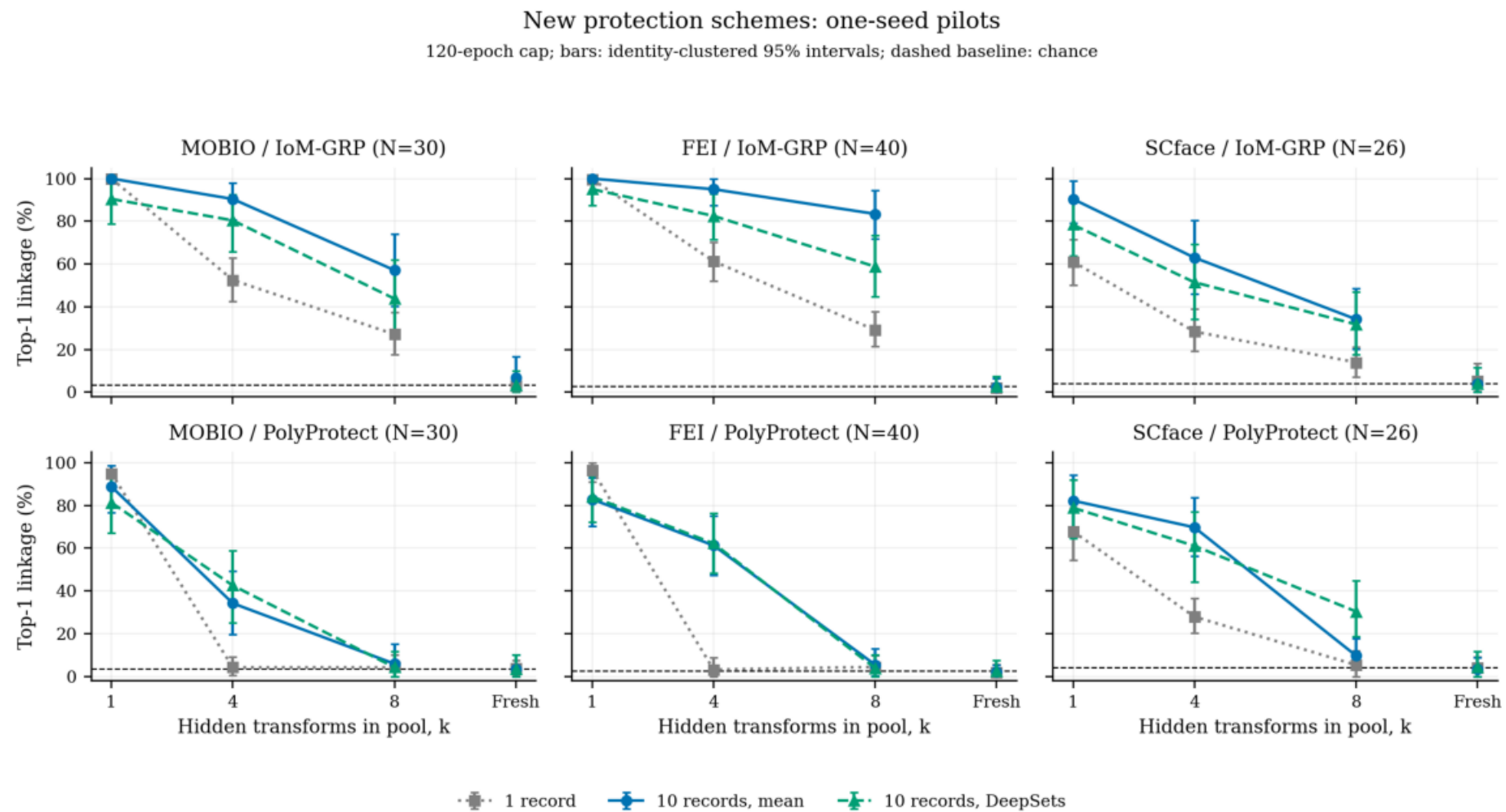
Mechanism and correlation controls

MOBIO mechanism controls (30 gallery identities; chance 3.33%)



Slot identifiers are not key values. Coarse and fine correlation sweeps use separate partitions.

Approved scheme pilots: one seed



Fresh PolyProtect native top-1: MOBIO 12.73%, FEI 13.73%, SCface 10.66%; separate diagnostic.

Ready to draft; not yet ready to submit

Completed: two added datasets, two new schemes, 24 pilot cells, paired intervals.

Before submission:

1. Freeze and authorize staged confirmation; one-seed pilots are not confirmation.
2. Decide whether AgeDB adds enough value for a third added dataset.
3. Approve statistical margins and a multiple-comparison plan.
4. Independently review the corrected theorem and related work.
5. Obtain Sani's scientific and presentation review.

A/A* is a venue ambition, not an established property or an acceptance prediction.